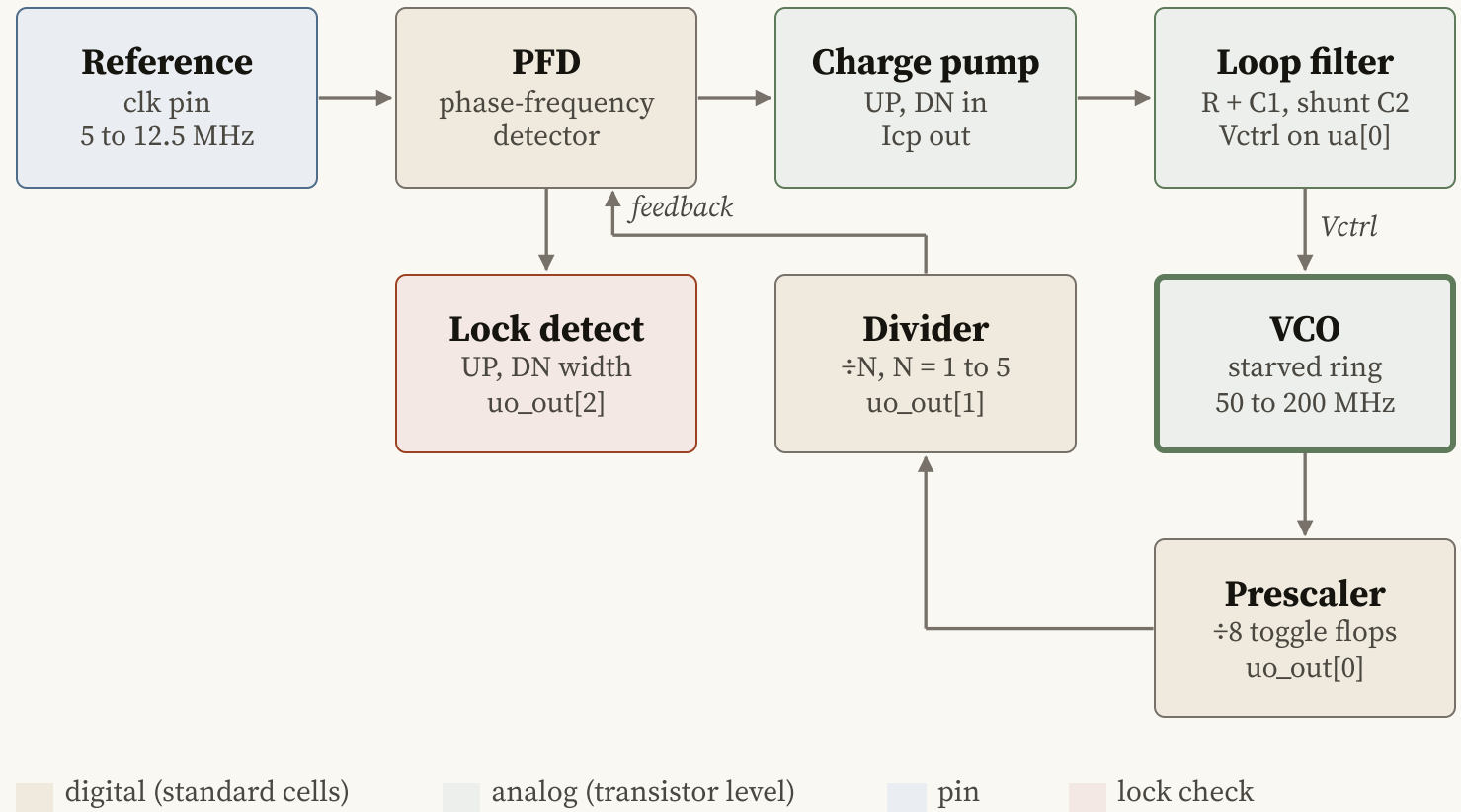


A charge-pump PLL on Tiny Tapeout GF180

Statement of purpose

- Build a classic charge-pump phase-locked loop: a clock multiplier, the standard circuit every chip uses to make a fast clock from a slow one.
- It multiplies a 5 to 12.5 MHz reference by $8N$ ($N = 1$ to 5) onto a 50 to 200 MHz on-chip oscillator.
- Mixed-signal by design: the phase detector and dividers are standard-cell logic, and the charge pump, loop filter and VCO are transistor-level analog.
- It exercises the whole open-source flow: RTL and CocoTB, ngspice across corners, layout with DRC and LVS.

System diagram



Pin assignment and proposed specification

IO pin assignment (Tiny Tapeout analog tile)

Pin	Dir	Use
clk	in	reference, 5 to 12.5 MHz
rst_n, ena	in	standard reset and enable
ui_in[2:0]	in	N, the divide ratio (1 to 5)
ui_in[5:3]	in	spare
ui_in[6]	in	PLL enable (powers down pump, VCO)
ui_in[7]	in	observe VCO $\div 8$ or $\div 16$
uo_out[0]	out	VCO $\div 8$ or $\div 16$
uo_out[1]	out	feedback divider output
uo_out[2]	out	lock
uo_out[4:3]	out	UP, DN (debug)
uo_out[7:5]	out	spare, driven low
uio[1:0]	in	charge-pump current trim
uio[7:2]	in	spare
ua[0]	analog	Vctrl, the loop-filter node
ua[1]	analog	bias reference (external resistor)

Proposed specification

Parameter	Target
Reference	5 to 12.5 MHz (nominal 6.25 MHz)
Multiplication	8N with N = 1 to 5, so 8 $\times$ to 40 $\times$
Nominal output	6.25 MHz $\times$ 16 = 100 MHz
VCO range	50 to 200 MHz, typical corner
Prescaler	fixed $\div 8$, so all logic runs $\leq$ 50 MHz
Charge pump	I _{cp} $\approx$ 20 μ A, 2-bit trim
Loop filter	R $\approx$ 26 k Ω , C1 $\approx$ 20 pF, C2 $\approx$ C1/10
Phase margin	$\geq 45^\circ$ at N = 2 (8N = 16)
Lock time	under 50 μ s
Observed output	VCO $\div 8$ or $\div 16$ on uo_out[0]

First ngspice runs at the typical corner: I_{cp} 19 μ A, VCO 10 to 258 MHz, phase margin 49.5 $^\circ$ at 8N = 16.

Timeline for completion and who does what

Timeline for completion

- **Oct 16** End of reading week: RTL for the PFD, prescaler, divider and lock detector with CocoTB passing; first charge pump and VCO schematics
- **Nov 5** Evaluation: each stage simulated in ngspice; the full loop locks in simulation at the typical corner
- **Nov 26** System verification: layout DRC and LVS clean, Tiny Tapeout precheck green
- **Dec 3** Documentation and a silicon test plan

Who does what

Ihsan Salari

DETECT AND FILTER

- PFD and charge pump
- Loop filter
- Design, simulation and layout of these stages

Dhyey Bhatt

OSCILLATE AND DIVIDE

- VCO
- Prescaler, $\div N$ divider and lock detector
- Design, simulation and layout of these stages

Both

TOGETHER

- Top-level integration and full-loop simulation
- Documentation, GitHub and presentations